



Advanced Time Series Analysis and Forecasting

Seminar 3: Structural VAR and local projections



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ AAA: highest credit rating (AAA-rated euro-area government bonds)
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ BIC: Bayesian Information Criterion
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BQ: Blanchard–Quah (long-run identification)
- ▣ CPI: Consumer Price Index
- ▣ EA: Euro Area
- ▣ EBP: Excess Bond Premium (Gilchrist–Zakrajšek)
- ▣ ECB: European Central Bank
- ▣ EIA: US Energy Information Administration
- ▣ FEVD: Forecast Error Variance Decomposition
- ▣ FF4: surprise in the federal funds futures contract three months ahead
- ▣ FOMC: Federal Open Market Committee
- ▣ FRED: Federal Reserve Economic Data
- ▣ GDP: Gross Domestic Product
- ▣ GNP: Gross National Product
- ▣ GNPC96: Real Gross National Product (FRED series code)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HICP: Harmonised Index of Consumer Prices
- ▣ IGERA: Index of Global Real Economic Activity (FRED series code)
- ▣ IP: Industrial Production
- ▣ LNS14000025: Unemployment rate, men aged 20 and over (FRED series code)
- ▣ LP: Local Projection
- ▣ LP-IV: Local Projection with Instrumental Variables

Acronyms used in this material (2/2)

- ▣ **MA: Moving Average**
- ▣ **MSE: Mean Squared Error**
- ▣ **OLS: Ordinary Least Squares**
- ▣ **ROBOR: Romanian Interbank Offer Rate**
- ▣ **SE: Standard Error**
- ▣ **SVAR: Structural Vector AutoRegression**
- ▣ **TSA: Time Series Analysis**
- ▣ **US: United States**
- ▣ **VAR: Vector AutoRegression**

Today's question and route

- **Question:** how do we turn the correlations of a VAR into the effects of identified shocks, and how much should we trust them?
 - ▶ this seminar comes **before** Lecture 3: the section “Prerequisites for Today” gives every definition the tasks use
- Route
 - ▶ Part A: Cholesky by hand, a known elasticity, a long-run restriction, the sign-restricted set, an external instrument, LP in an AR(1), FEVD, identification through heteroskedasticity
 - ▶ Part B: Kilian (2009) on oil; a Romanian VAR and its orderings; Gertler and Karadi (2015); Romer–Romer shocks by LP and VAR; Blanchard–Quah robustness
 - ▶ Part C: euro-area spillovers to Romania with a weak instrument; an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	Cholesky by hand; a non-recursive system with a known elasticity	Solved, Proposed	A1
A3, A4	a long-run restriction; the set identified by signs	Solved, Proposed	A3
A5, A6	an external instrument; local projections in an AR(1)	Solved, Proposed	A5
A7, A8	variance decomposition; identification through heteroskedasticity	Solved, Proposed	A7
B1, B2	Kilian (2009) on oil; orderings of a Romanian VAR	Solved, Proposed	B1
B3–B5	Gertler–Karadi proxy SVAR; Romer–Romer by LP and VAR; Blanchard–Quah robustness	Solved, Proposed	B1, B3
C1, C2	euro-area spillovers with a weak instrument; what is wrong in an AI answer?	Proposed	B3, A1–A8

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
oil production, real activity, real oil price	EIA, FRED	monthly	1974–2026
RO and EA IP, HICP, 3-month rates; EUR/RON	Eurostat, BNR	monthly	2005–2026
US macro data, FF4, Romer–Romer shocks	Ramey (2016) files	monthly	1965–2015
real GNP, unemployment of men 20+	FRED	quarterly	1948–2026
ECB policy shocks; EA 1-year rate	Jarociński–Karadi; ECB	monthly	2005–2025

- In the notebook: `var_ols`, `irf`, `fevd`, `bq_impact`, `bootstrap_var`, `proxy_irf`, `proxy_mbb`, `lp`, `lp_iv`; no account or key is needed

Prerequisites for Today (1/6): reduced and structural form

- Reduced form (TSA, Chapter 6): $y_t = \nu + A_1 y_{t-1} + \dots + A_p y_{t-p} + u_t$, $\Sigma_u = \mathbb{E} u_t u_t'$
 - ▶ y_t : n variables; ν : intercepts; A_j : $n \times n$ coefficient matrices; u_t : one-step forecast errors with covariance Σ_u
- Structural form: $u_t = B_0 \varepsilon_t$, $\mathbb{E} \varepsilon_t \varepsilon_t' = I$
 - ▶ ε_t : uncorrelated structural shocks of unit variance; B_0 : their impact effects (column j for shock j)
- Impulse responses: $\Theta_h = \Phi_h B_0$
 - ▶ $\Phi_0 = I$, $\Phi_h = \sum_{j=1}^{\min(h,p)} A_j \Phi_{h-j}$: the reduced-form MA coefficients; $\theta_{ij,h}$: the response of y_i to shock j after h periods

Prerequisites for Today (2/6): identification, Cholesky, FEVD

- $\Sigma_u = B_0 B_0'$ gives $n(n+1)/2$ equations for n^2 unknowns: $n(n-1)/2$ restrictions are needed
 - ▶ $B_0 Q$ with Q orthogonal ($QQ' = I$) fits the data equally well (observational equivalence)
- **Cholesky**: $B_0 = P$, lower triangular, $PP' = \Sigma_u$
 - ▶ for 2×2 : $p_{11} = \sqrt{\sigma_{11}}$, $p_{21} = \sigma_{21}/p_{11}$, $p_{22} = \sqrt{\sigma_{22} - p_{21}^2}$; σ_{ij} : the elements of Σ_u
 - ▶ the first variable does not react within the period to the second shock (a timing assumption)
- **FEVD** (forecast error variance decomposition): the share of shock j in the h -step forecast error variance of y_i
 - ▶ $\sum_{s < h} \theta_{ij,s}^2 / \sum_{s < h} \sum_k \theta_{ik,s}^2$

Prerequisites for Today (3/6): long run, signs, heteroskedasticity

- **Long run** Blanchard and Quah (1989): $\Theta(1) = \sum_h \Theta_h = A(1)^{-1}B_0$, $A(1) = I - \sum_j A_j$
 - ▶ $\Theta(1)$: the cumulative (long-run) responses; a zero in $\Theta(1)$ means a shock with no permanent effect
 - ▶ $\Theta(1) = \text{chol}(A(1)^{-1}\Sigma_u A(1)^{-1'})$ (chol: the lower-triangular Cholesky factor), then $B_0 = A(1)\Theta(1)$
- **Signs** Uhlig (2005), Rubio-Ramírez, Waggoner and Zha (2010): $B_0 = PQ$; keep the Q whose responses have the required signs
 - ▶ in 2×2 , $Q = R(\theta)$, a rotation by angle θ ; the result is an identified set, not a point
- **Heteroskedasticity** Rigobon (2003): two regimes with $\Sigma_1 = B_0 B_0'$, $\Sigma_2 = B_0 \Lambda B_0'$
 - ▶ Λ : a diagonal matrix of variance ratios λ_i ; they are the eigenvalues of $\Sigma_1^{-1}\Sigma_2$

Prerequisites for Today (4/6): external instruments

- **Proxy** z_t [Stock and Watson \(2018\)](#), [Mertens and Ravn \(2013\)](#): an observed variable related only to shock 1
 - ▶ relevance $\mathbb{E}z_t\varepsilon_{1t} \neq 0$; exogeneity $\mathbb{E}z_t\varepsilon_{jt} = 0$ for $j \neq 1$
 - ▶ then $\mathbb{E}u_t z_t \propto b_1$ (the first column of B_0): relative impacts $b_{i1}/b_{11} = \text{Cov}(u_{it}, z_t)/\text{Cov}(u_{1t}, z_t)$
- **First stage**: $u_{1t} = \pi z_t + e_t$
 - ▶ $F = (\hat{\pi}/\text{se})^2$ with robust SE; $F < 10$ signals a weak instrument
- **High-frequency surprises**: FF4 = change of the three-month-ahead federal funds futures rate in a 30-minute window around an FOMC announcement [Gertler and Karadi \(2015\)](#)
 - ▶ ECB analogue: the pure monetary policy shocks of [Jarociński and Karadi \(2020\)](#)

Prerequisites for Today (5/6): weak instruments and local projections

- **Anderson–Rubin set** [Montiel Olea, Stock and Watson \(2021\)](#) for the effect β of x on y with instrument z
 - ▶ all β for which $y - \beta x$ is unrelated to z (the coefficient of z is not significant); valid with weak instruments, possibly unbounded
- **LP** [Jordà \(2005\)](#): one regression per horizon h
 - ▶ $y_{t+h} = \mu_h + \beta_h x_t + \gamma_h' w_t + \xi_{t+h}$
 - ▶ x_t : the shock; w_t : controls (lags); $\hat{\beta}_h$: the response at horizon h ; ξ_{t+h} : the error
 - ▶ ξ_{t+h} is autocorrelated: Newey–West with $h + 1$ lags, or one extra lag of the controls and robust SE (lag augmentation) [Montiel Olea and Plagborg-Møller \(2021\)](#)
- **LP-IV**: x_t = the policy variable instrumented by z_t [Stock and Watson \(2018\)](#)
 - ▶ with the same lags, LP and VAR agree up to horizon p [Plagborg-Møller and Wolf \(2021\)](#)

Prerequisites for Today (6/6): inference for responses

- ▣ **Bootstrap** for VAR responses [Gonçalves and Kilian \(2004\)](#), [Jentsch and Lunsford \(2019\)](#)
 - ▶ residual (i.i.d. errors); wild ($\pm \hat{u}_t$, conditional heteroskedasticity); moving block (proxy SVAR)
 - ▶ percentile bands at 68%, 90% or 95%, pointwise in h
- ▣ **Delta method** for a smooth function g of an estimate $\hat{\rho}$:
 - ▶ $\text{se}(g(\hat{\rho})) \approx |g'(\rho)| \text{se}(\hat{\rho})$
 - ▶ g' : the derivative; e.g. $g(\rho) = \rho^4$ gives $\text{se}(\hat{\rho}^4) \approx 4\rho^3 \text{se}(\hat{\rho})$

A1: Cholesky by hand [Solved]

Task

- ▣ Bivariate VAR(1), $A_1 = \begin{pmatrix} 0.5 & 0 \\ 0.2 & 0.4 \end{pmatrix}$, $\Sigma_u = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix}$.
- ▣ 1. Count the moments, the unknowns of B_0 and the restrictions needed.
- ▣ 2. Compute the Cholesky factor P and state the timing assumption it encodes.
- ▣ 3. Compute the responses $\Theta_1 = A_1 P$ after one period.
- ▣ Report: three counts, one matrix, one sentence, one matrix.

Solution

- ▣ 1. Moments $n(n+1)/2 = 3$, unknowns $n^2 = 4$, restrictions $n(n-1)/2 = 1$
- ▣ 2. $p_{11} = 2$, $p_{21} = 1/2 = 0.5$, $p_{22} = \sqrt{2 - 0.25}$:
 $P = \begin{pmatrix} 2.000 & 0.000 \\ 0.500 & 1.323 \end{pmatrix}$; variable 1 does not react to shock 2 within the period
- ▣ 3. $\Theta_1 = A_1 P = \begin{pmatrix} 1.000 & 0.000 \\ 0.600 & 0.529 \end{pmatrix}$: shock 1 still moves variable 2 after one period through a_{21}

A2: a non-recursive system with a known elasticity [Proposed]

Solution discussed in class

Task

- $u_1 = a_{12}u_2 + \sigma_1\varepsilon_1$ (policy reacts to output), $u_2 = a_{21}u_1 + \sigma_2\varepsilon_2$ with $a_{21} = -0.2$ known (as in Blanchard and Perotti (2002)); Σ_u of A1. Model: A1.
- 1. Show that $e_2 = u_2 - a_{21}u_1$ is a valid instrument for u_2 in the first equation.
- 2. Compute $a_{12} = \text{Cov}(u_1, e_2)/\text{Cov}(u_2, e_2)$, then σ_1^2 and σ_2^2 .
- 3. Write B_0 and check $B_0B_0' = \Sigma_u$; explain why this scheme is not recursive.
- Report: one argument, three numbers, one matrix.

A3: a long-run restriction by hand [Solved]

Task

- ▣ $y_t = (\Delta \text{output}_t, u_t)'$, VAR(1) with $A_1 = \begin{pmatrix} 0.2 & -0.3 \\ 0.1 & 0.6 \end{pmatrix}$,
 $\Sigma_u = \begin{pmatrix} 1 & -0.2 \\ -0.2 & 0.5 \end{pmatrix}$.
- ▣ 1. Compute $A(1)^{-1} = (I - A_1)^{-1}$ and $A(1)^{-1}\Sigma_u A(1)^{-1}$.
- ▣ 2. Impose that shock 2 (demand) has no long-run effect on output and compute $\Theta(1)$ and B_0 .
- ▣ 3. Read the sign of the impact of the supply shock on unemployment.
- ▣ Report: two matrices, two matrices, one sentence.

Solution

- ▣ 1. $\det(I - A_1) = 0.8 \cdot 0.4 + 0.03 = 0.35$;
 $A(1)^{-1} = \begin{pmatrix} 1.143 & -0.857 \\ 0.286 & 2.286 \end{pmatrix}$; product = $\begin{pmatrix} 2.065 & -1.127 \\ -1.127 & 2.433 \end{pmatrix}$
- ▣ 2. $\Theta(1) = \text{chol}(\cdot) = \begin{pmatrix} 1.437 & 0.000 \\ -0.784 & 1.348 \end{pmatrix}$;
 $B_0 = A(1)\Theta(1) = \begin{pmatrix} 0.915 & 0.405 \\ -0.457 & 0.539 \end{pmatrix}$
- ▣ 3. The supply shock lowers unemployment on impact here; in the US data of BQ it raises it at first (Lecture 3)

A4: the set identified by sign restrictions [Proposed]

Solution discussed in class

Task

- ▣ Price and quantity innovations, $\Sigma_u = \begin{pmatrix} 1 & 0.3 \\ 0.3 & 0.5 \end{pmatrix}$; $B_0 = PR(\theta)$, first column $P(\cos \theta, \sin \theta)'$. Demand: price +, quantity +; supply: opposite signs. Model: A3.
- ▣ 1. Compute P and write the two impact responses to the demand shock as functions of θ .
- ▣ 2. Find the interval of θ for which all four signs hold.
- ▣ 3. Give the identified set of the impact effect of demand on quantity, and explain why a uniform prior on θ is not neutral.
- ▣ Report: one matrix, one interval, one interval, one sentence.

A5: an external instrument by hand [Solved]

Task

- ▣ Residuals of (policy rate, output, prices); $\text{Cov}(u_t, z_t) = (0.020, -0.006, 0.010)'$; first stage of u_{1t} on z_t : $\hat{\pi} = 0.30$, robust SE 0.08.
- ▣ 1. Compute the relative impacts b_{11}/b_{11} and the impacts of a shock that raises the rate by 25 bp.
- ▣ 2. Compute the first-stage t and F and judge relevance.
- ▣ 3. Name one way the exogeneity condition could fail for an FOMC surprise.
- ▣ Report: four numbers, two statistics, one sentence.

Solution

- ▣ 1. $b_{21}/b_{11} = -0.006/0.020 = -0.30$, $b_{31}/b_{11} = 0.50$; 25 bp: output -0.075 , prices 0.125
- ▣ 2. $t = 0.30/0.08 = 3.75$, $F = t^2 = 14.1 > 10$: relevant by the rule of thumb
- ▣ 3. The information effect [Nakamura and Steinsson \(2018\)](#): the surprise also reveals the Fed's view of the economy, so $\mathbb{E}z_t \varepsilon_{jt} \neq 0$ for a demand shock j ; prices rising after a tightening (here $+$) is a warning sign

A6: local projections in an AR(1) [Proposed]

Solution discussed in class

Task

- ▣ $y_t = \rho y_{t-1} + \varepsilon_t$, $\rho = 0.9$, $\text{Var}\varepsilon_t = 1$, $T = 200$; the response at horizon $h = 4$. Model: A5.
- ▣ 1. Show that the LP of y_{t+4} on y_t has population coefficient ρ^4 and error $\sum_{j=0}^3 \rho^j \varepsilon_{t+4-j}$.
- ▣ 2. Give the MA order and the variance of the LP error, and say why HAC errors are needed.
- ▣ 3. With $\text{se}(\hat{\rho}) = \sqrt{(1 - \rho^2)/T}$, give the delta-method SE of the plug-in $\hat{\rho}^4$ and explain why LP is noisier.
- ▣ Report: one derivation, two numbers, two numbers, one sentence.

A7: variance decomposition by hand [Solved]

Task

- ▣ The VAR(1) and the Cholesky factor P of A_1 ; $\Theta_0 = P$, $\Theta_1 = A_1 P$, $\Theta_2 = A_1^2 P$.
- ▣ 1. Compute the share of shock 1 in the one-step forecast error variance of variable 2.
- ▣ 2. Repeat for two and three steps.
- ▣ 3. Explain why the share of shock 2 in variable 1 stays zero at every horizon.
- ▣ Report: three shares, one sentence.

Solution

- ▣ 1. $h = 1$: $0.5^2 / (0.5^2 + 1.75) = 12.5\%$
- ▣ 2. $\Theta_1 = \begin{pmatrix} 1.000 & 0.000 \\ 0.600 & 0.529 \end{pmatrix}$; $h = 2$:
 $(0.25 + 0.36) / (0.25 + 1.75 + 0.36 + 0.28) = 23.1\%$; $h = 3$: 27.9%
- ▣ 3. $p_{12} = 0$ and $a_{12} = 0$: variable 1 never responds to shock 2, so its forecast error contains only shock 1

A8: identification through heteroskedasticity [Proposed]

Solution discussed in class

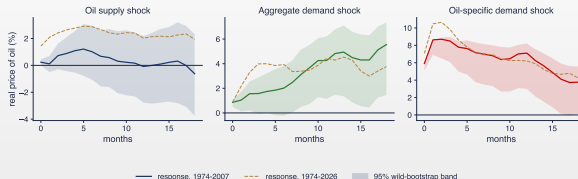
Task

- ▣ Two regimes: $\Sigma_1 = \begin{pmatrix} 1.25 & 0.80 \\ 0.80 & 1.09 \end{pmatrix}$, $\Sigma_2 = \begin{pmatrix} 4.25 & 1.70 \\ 1.70 & 1.36 \end{pmatrix}$; B_0 is the same, the variance of shock 1 changes. Model: A7.
- ▣ 1. Show that $\Sigma_1^{-1}\Sigma_2 = B_0^{-1'}\Lambda B_0'$ and compute its eigenvalues.
- ▣ 2. Recover B_0 (normalise $b_{11} = b_{22} = 1$) from $\Sigma_2 - \Sigma_1$.
- ▣ 3. State the two assumptions that make this work and the one that fails if both variances change by the same factor.
- ▣ Report: two eigenvalues, one matrix, three sentences.

B1: oil supply and demand shocks [Solved]

- **Question:** does the real price of oil respond differently to supply, aggregate demand and oil-specific demand shocks, as in Kilian (2009)?
- **Inputs:** percent change of world crude production (EIA), real activity (FRED IGBREA), log real refiner acquisition cost (EIA, deflated by CPI); monthly 1974–2007 and 1974–2026; VAR(24), Cholesky in this order
- **Tasks**
 1. Estimate the VAR(24) on 1974–2007 and identify the three shocks, signing the supply shock so that it raises the price.
 2. Report the impact and 12-month responses of the real price with 95% recursive-design wild-bootstrap bands.
 3. Compute the 60-month FEVD of the price on both samples.
 4. Interpretation: which shock moves the oil price, and did the answer change after 2007?
- **Report:** six responses with bands, six variance shares, one sentence
- **Notebook:** section B1

B1: solution [Solved]



Shock	impact	12 months	95% band	FEVD 60, 2007	FEVD 60, 2026
oil supply	0.24%	-0.1%	[-2.8, 2.2]	1%	9%
aggregate demand	0.86%	4.8%	[1.5, 6.8]	64%	31%
oil-specific demand	5.90%	7.1%	[3.0, 8.2]	35%	60%

- Interpretation: demand moves the price (aggregate demand with a delay, oil-specific demand at once), supply hardly at all before 2007; on the longer sample supply shocks matter more (12-month effect 2.0%) and oil-specific demand explains most of the variance

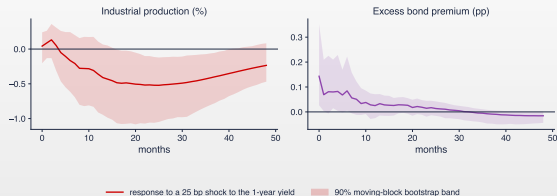
B2: orderings of a Romanian VAR [Proposed]

- **Question:** is the rise of Romanian prices after a ROBOR shock (the price puzzle) an artefact of the ordering or of the lag length?
- **Inputs:** monthly 2005–2026: EA IP, EA HICP, Euribor 3M, RO IP, RO HICP, ROBOR 3M, EUR/RON (Eurostat, BNR), logs $\times 100$, monthly dummies; model: B1
- **Tasks**
 1. Select the lag length by BIC and AIC.
 2. For each lag length, estimate the VAR with ROBOR after RO prices, ROBOR first in the RO block and ROBOR last, keeping the euro area first.
 3. Report the 12- and 24-month responses of RO HICP, and the 12-month responses of EUR/RON and RO IP, to a 25 bp ROBOR shock.
 4. Interpretation: what does the stability of the puzzle across orderings tell you about the identification?
- **Report:** two lag lengths, a table of six rows, one sentence
- **Notebook:** section B2

B3: replicating Gertler and Karadi (2015) [Solved]

- **Question:** does a monetary policy shock identified with FF4 surprises raise credit spreads and lower output?
- **Inputs:** 1-year Treasury yield, log IP, log CPI, excess bond premium; VAR(12), 1979:7–2012:6; instrument FF4, 1991:1–2012:6 (replication files of [Ramey \(2016\)](#))
- **Tasks**
 1. Estimate the reduced-form VAR and regress the yield residual on FF4 over the overlap: report the robust F .
 2. Compute the impact column from $\text{Cov}(u_t, z_t)$, scaled to a 25 bp rise in the yield, and the responses for 48 months.
 3. Build 90% moving-block bootstrap bands for IP and the excess bond premium.
 4. Interpretation: through which channel does the shock work, according to these responses?
- **Report:** one F , four impacts, the IP trough with its band, one sentence
- **Notebook:** section B3

B3: solution [Solved]



- First stage on 258 months: $\hat{\pi} = 1.16$, robust $F = 17.7$ (homoskedastic 21.9); impact: yield 0.250, IP 0.042%, CPI -0.042% , EBP 0.143 pp (band $[0.03, 0.35]$)
- IP reaches -0.52% after 25 months (band $[-1.05, -0.12]$)
- Interpretation: the credit channel: the bond premium jumps at once, then output falls with a lag; the instrument is relevant ($F > 10$), as in the paper

B4: Romer–Romer shocks by LP and by VAR [Proposed]

- **Question:** do local projections and a VAR with the same narrative shock give the same output and price responses?
- **Inputs:** Romer–Romer shocks updated by Wieland (data of [Ramey \(2016\)](#)), log IP, unemployment, log CPI, log commodity prices, funds rate; monthly 1969:3–2007:12 ($T = 466$); model: B3
- **Tasks**
 1. Estimate LP of IP, CPI and the funds rate on the shock for $h = 0, \dots, 48$, with 2 lags of all variables and of the shock, Newey–West SE with $h + 1$ lags.
 2. Estimate a VAR(12) with the shock ordered first and scale its responses to a 1 pp shock.
 3. Compare the trough of IP, the CPI response after 48 months and the impact on the funds rate.
 4. Interpretation: which estimator would you report, and why?
- **Report:** three pairs of numbers, two SE, one sentence
- **Notebook:** section B4

B5: how robust is Blanchard–Quah? [Proposed]

- **Question:** does the share of demand shocks in output fluctuations survive changes in the treatment of low frequencies?
- **Inputs:** real GNP growth (FRED GNPC96), unemployment of men 20+ (LNS14000025), quarterly, VAR(8); model: B1
- **Tasks**
 1. Estimate the BQ model with the original treatment (break in mean growth at 1973:4, linear trend in unemployment), 1948–1987.
 2. Re-estimate with no break and no trend, and with the original treatment on data up to 2019.
 3. Report the share of demand shocks in the output-level FEVD at 4 and 40 quarters, and the long-run effect of a supply shock.
 4. Interpretation: what does the comparison say about long-run restrictions?
- **Report:** a table of three rows and four columns, one sentence
- **Notebook:** section B5

C1: euro-area spillovers with a weak instrument [Proposed]

- **Question:** how much does Romanian HICP move 12 months after a 1 pp rise of the euro-area 1-year rate caused by an ECB policy shock?
- **Inputs:** Romanian data of B2, euro-area 1-year AAA spot rate (ECB), pure policy shocks of [Jarociński and Karadi \(2020\)](#); origins 2005:8–2025:10; model: B3
- **Tasks**
 1. Write a five-line pre-registration: outcome, horizon, instrument, controls, inference.
 2. Estimate the LP-IV at $h = 12$ with 3 lags of all variables and of the shock and monthly dummies; report $\hat{\beta}$, its Newey–West SE and the first-stage F .
 3. Build the 90% Anderson–Rubin set on a grid of $\beta \in [-15, 15]$.
 4. Interpretation: what can and cannot be concluded about ECB spillovers to Romanian prices?
- **Report:** a five-line plan, one estimate with SE, one F , one set, a project plan
- Notebook: section C1

C2: audit an AI answer [Proposed]

- A student asked an AI assistant to summarise structural VAR identification. The answer:
- (a) A four-variable SVAR needs four restrictions to be exactly identified.
- (b) In a Cholesky ordering, reordering the variables placed after the policy variable does not change the responses to the policy shock.
- (c) With enough horizons restricted, sign restrictions point-identify the output response.
- (d) In a proxy SVAR, b_{21}/b_{11} equals $\text{Cov}(u_2, z)/\text{Cov}(u_1, z)$.
- (e) Local projections are unbiased and have smaller variance than VARs, so they always dominate.
- (f) The wild bootstrap of Mertens and Ravn gives valid bands for proxy SVARs.
- (g) With $\Sigma = \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}$, the Cholesky impact of shock 1 on variable 2 is 1.
- Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
 - ▶ 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- ▣ Every structural answer rests on an identifying assumption: count it, state it, defend it
- ▣ Cholesky, long-run zeros and known elasticities give points; signs give sets
- ▣ An external instrument identifies one shock if it is relevant and exogenous; check F and use robust sets
- ▣ LP and VAR answer the same question with different bias and variance
- ▣ An AI answer is a draft: check the counts, the sets, the bootstrap and the arithmetic

After the seminar

- ▣ Lecture 3 develops each topic of today: identification schemes, set identification, external instruments, inference, local projections, Romania
- ▣ Try the [Proposed] tasks in the notebook
- ▣ C1 can grow into a team project: ECB spillovers to Romania with high-frequency instruments
- ▣ Reading: [Kilian \(2009\)](#); [Gertler and Karadi \(2015\)](#); [Stock and Watson \(2018\)](#); [Plagborg-Møller and Wolf \(2021\)](#)
- ▣ **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

References (1/2)

- Baumeister, C., & Hamilton, J. D. (2015). Sign restrictions, structural vector autoregressions, and useful prior information. *Econometrica*, 83(5), 1963–1999. doi:10.3982/ECTA12356
- Blanchard, O. J., & Quah, D. (1989). The dynamic effects of aggregate demand and supply disturbances. *American Economic Review*, 79(4), 655–673. jstor.org
- Blanchard, O., & Perotti, R. (2002). An empirical characterization of the dynamic effects of changes in government spending and taxes on output. *The Quarterly Journal of Economics*, 117(4), 1329–1368. doi:10.1162/003355302320935043
- Christiano, L. J., Eichenbaum, M., & Evans, C. L. (1999). Monetary policy shocks: What have we learned and to what end? In J. B. Taylor & M. Woodford (Eds.), *Handbook of Macroeconomics* (Vol. 1A, pp. 65–148). Elsevier. doi:10.1016/S1574-0048(99)01005-8
- Faust, J., & Leeper, E. M. (1997). When do long-run identifying restrictions give reliable results? *Journal of Business & Economic Statistics*, 15(3), 345–353. doi:10.1080/07350015.1997.10524712
- Gertler, M., & Karadi, P. (2015). Monetary policy surprises, credit costs, and economic activity. *American Economic Journal: Macroeconomics*, 7(1), 44–76. doi:10.1257/mac.20130329
- Gonçalves, S., & Kilian, L. (2004). Bootstrapping autoregressions with conditional heteroskedasticity of unknown form. *Journal of Econometrics*, 123(1), 89–120. doi:10.1016/j.jeconom.2003.10.030
- Jarociński, M., & Karadi, P. (2020). Deconstructing monetary policy surprises: The role of information shocks. *American Economic Journal: Macroeconomics*, 12(2), 1–43. doi:10.1257/mac.20180090
- Jentsch, C., & Lunsford, K. G. (2019). The dynamic effects of personal and corporate income tax changes in the United States: Comment. *American Economic Review*, 109(7), 2655–2678. doi:10.1257/aer.20162011
- Jordà, Ò. (2005). Estimation and inference of impulse responses by local projections. *American Economic Review*, 95(1), 161–182. doi:10.1257/0002828053828518
- Kilian, L. (2009). Not all oil price shocks are alike: Disentangling demand and supply shocks in the crude oil market. *American Economic Review*, 99(3), 1053–1069. doi:10.1257/aer.99.3.1053
- Li, D., Plagborg-Møller, M., & Wolf, C. K. (2024). Local projections vs. VARs: Lessons from thousands of DGPs. *Journal of Econometrics*, 244(2), 105722. doi:10.1016/j.jeconom.2024.105722

References (2/2)

- ▣ Mertens, K., & Ravn, M. O. (2013). The dynamic effects of personal and corporate income tax changes in the United States. *American Economic Review*, 103(4), 1212–1247. doi:[10.1257/aer.103.4.1212](https://doi.org/10.1257/aer.103.4.1212)
- ▣ Montiel Olea, J. L., & Plagborg-Møller, M. (2021). Local projection inference is simpler and more robust than you think. *Econometrica*, 89(4), 1789–1823. doi:[10.3982/ECTA18756](https://doi.org/10.3982/ECTA18756)
- ▣ Montiel Olea, J. L., Stock, J. H., & Watson, M. W. (2021). Inference in structural vector autoregressions identified with an external instrument. *Journal of Econometrics*, 225(1), 74–87. doi:[10.1016/j.jeconom.2020.05.014](https://doi.org/10.1016/j.jeconom.2020.05.014)
- ▣ Nakamura, E., & Steinsson, J. (2018). High-frequency identification of monetary non-neutrality: The information effect. *The Quarterly Journal of Economics*, 133(3), 1283–1330. doi:[10.1093/qje/qjy004](https://doi.org/10.1093/qje/qjy004)
- ▣ Plagborg-Møller, M., & Wolf, C. K. (2021). Local projections and VARs estimate the same impulse responses. *Econometrica*, 89(2), 955–980. doi:[10.3982/ECTA17813](https://doi.org/10.3982/ECTA17813)
- ▣ Ramey, V. A. (2016). Macroeconomic shocks and their propagation. In J. B. Taylor & H. Uhlig (Eds.), *Handbook of Macroeconomics* (Vol. 2A, pp. 71–162). Elsevier. doi:[10.1016/bs.hesmac.2016.03.003](https://doi.org/10.1016/bs.hesmac.2016.03.003)
- ▣ Rigobon, R. (2003). Identification through heteroskedasticity. *The Review of Economics and Statistics*, 85(4), 777–792. doi:[10.1162/003465303772815727](https://doi.org/10.1162/003465303772815727)
- ▣ Rubio-Ramírez, J. F., Waggoner, D. F., & Zha, T. (2010). Structural vector autoregressions: Theory of identification and algorithms for inference. *The Review of Economic Studies*, 77(2), 665–696. doi:[10.1111/j.1467-937X.2009.00578.x](https://doi.org/10.1111/j.1467-937X.2009.00578.x)
- ▣ Stock, J. H., & Watson, M. W. (2018). Identification and estimation of dynamic causal effects in macroeconomics using external instruments. *The Economic Journal*, 128(610), 917–948. doi:[10.1111/ecoj.12593](https://doi.org/10.1111/ecoj.12593)
- ▣ Uhlig, H. (2005). What are the effects of monetary policy on output? Results from an agnostic identification procedure. *Journal of Monetary Economics*, 52(2), 381–419. doi:[10.1016/j.jmoneco.2004.05.007](https://doi.org/10.1016/j.jmoneco.2004.05.007)